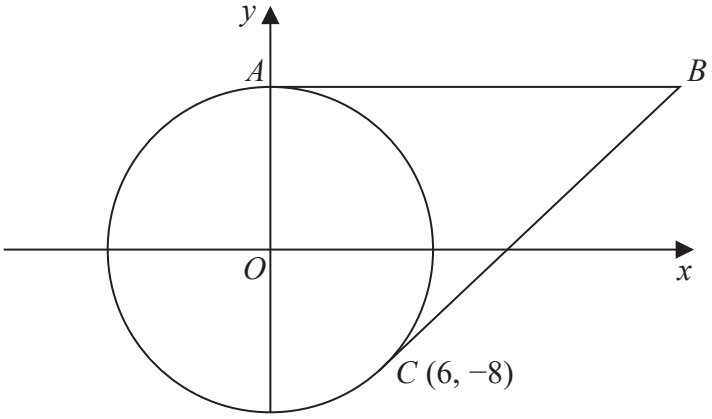
21



The diagram shows the circle with equation $x^2 + y^2 = 100$
The unit of length on both axes is one centimetre.

The circle intersects the positive y-axis at the point A.
The point C on the circle has coordinates $(6, -8)$
The straight lines AB and CB are tangents to the circle.

Find the area of quadrilateral $ABCO$.

Radius = 10 cm, so A = (0,10). Let AB = z.

AB = BC = z: tangents from B have equal length.

1 mark
Equal tangents.

B = (z,10); vertical distance B to C = 18.

By Pythagoras in the coordinate triangle:

$BC^2 = (z - 6)^2 + 18^2$

1 mark
Use Pythagoras.

$z^2 = z^2 - 12z + 36 + 324$

$12z = 360$, so $z = 30$ cm.

1 mark
Solve for tangent length.

OA and OC are perpendicular to their tangents.

Area ABCO = ½×10×30 + ½×10×30 = 300 cm^2.

1 mark
Two right triangle areas.

..... cm^2

(Total for Question 21 is 4 marks)